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1.a Number of blocks in main memory = $\frac{2^{32}}{2^5} = 2^{27}$ blocks

b. Number of block in cache = $\frac{2^{15}}{2^5} = 2^{10}$ blocks

c. Direct-mapped

Size of offset = $2^5 = 32$ bytes
= 5 bits

Index field = $2^{10} = 10$ bits

$\therefore$ tag field = $32 - (10 + 5)$
= 17 bits

d. fully-associative

offset field = $2^5 = 5$ bits

$\therefore$ tag field = $32 - 5$

= 27 bits

e. 4-way set associative

offset field = $2^5 = 5$ bits

number of sets = $\frac{2^{10}}{2^2} = 2^8$ index bits

tag bits = $32 - (8 + 5)$
= 19 bits

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2. a. offset bit = $\log_2 16$

= 4 bit

index bit = $\log_2 8$

= 3 bit

$\therefore$ tag bit = $16 - (4 + 3)$

= $16 - 7$

= 9 bit

b.

Memory address	Tag	Index	Hit / Miss
0010 1101 1011 0011	0010 1101	011	Miss
0000 0110 1111 1100	0000 0110	111	Miss
0010 1101 1011 1000	0010 1101	011	Hit
1010 1010 1010 1011	1010 1010	010	Miss

c. Average access time = $(0.95 * 15) + (0.05 * (15 + 220))$

= $14.25 + 11.75$

= 26 ns.

3. 1. offset = 16 byte = 2^4 byte = 4

cache size = 4096×2^4

= $2^{12} \times 2^4$

= 2^{16}



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a. Direct-mapped

$$\begin{aligned} \text{tag-bit} &= 64 - 16 \rightarrow \text{index} = 16 - 4 \\ &= 48 \text{ bit} \end{aligned}$$

b. 2-way set associative

$$\begin{aligned} \text{tag-bit} &= 64 - (11 + 4) \\ &= 49 \text{ bit} \end{aligned}$$

$$\begin{aligned} \text{Total sets} &= 4096 / 2 \\ &= 2048 \\ &= 2^{11} \text{ bit} \end{aligned}$$

c. 4-way set associative

$$\begin{aligned} \text{tag-bit} &= 64 - (10 + 4) \\ &= 50 \text{ bit} \end{aligned}$$

$$\begin{aligned} \text{Total sets} &= 4096 / 4 \\ &= 1024 \\ &= 2^{10} \text{ bit} \end{aligned}$$

d. fully associative

$$\begin{aligned} \text{tag-bit} &= 64 - 4 \\ &= 60 \text{ bit} \end{aligned}$$

$$2. \text{Ratio} = (\text{tag bit} + \text{valid bit} + \text{data bit}) / \text{data bits}$$

a. Direct-mapped

$$(48 + 1 + (16 * 8)) / (16 * 8) = 177 / 128 = 1.38$$

b. 2-way associative

$$(49 + 1 + (16 * 8)) / (16 * 8) = 178 / 128 = 1.39$$

c. 4-way associative

$$(50 + 1 + (16 * 8)) / (16 * 8) = 179 / 128 = 1.398$$

d. fully associative

$$(60 + 1 + (16 * 8)) / (16 * 8) = 189 / 128 = 1.4$$

4. Page size = 64 byte

$$\text{offset} = \log_2(64) = 6 \text{ bits}$$

$$\text{virtual page} = 8$$

$$\text{no. of virtual page} = \log_2(8) = 3 \text{ bits}$$

$$\text{page frame number} = \log_2(4) = 2 \text{ bits}$$

a. bits in virtual address = $3 + 6 = 9 \text{ bits}$

b. bits in physical address = $2 + 6 = 8 \text{ bits}$

c. bits are virtual page number = 3 bits



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d. bits are page frame number = 2 bits

e. bits are page offset = 6 bits

f. 1. 0x00

Physical address = 01000000
= 0x40

2. 0x44

Physical address = 11000000
= 0xC0

3. 0xC2

physical address = 00000010
= 0x02

4. 0x80

Physical address = no physical address because of page fault.

5. a. No. of pages in virtual memory = $2^{10} / 2^4 = 2^6$ pages

b. No. of pages in physical memory = $2^8 / 2^4 = 2^4$ frames

c. Offset = $\log_2(2^4) = 4$ bits, total no. of bits = 10 bits

Virtual address = 10-4
= 6 bits

total no. of page = 2^6 entries.
table entries